

How to instal TwinCAT

- TwinCAT is Windows Software implemented in Microsoft Visual Studio
- Use a remote desktop if you lack a Windows computer and want to work at home.
 - Obs! A good internet connection is needed
- All computers at University West have TwinCAT already installed

Install TwinCAT on your Windows Computer



- Click on this link <https://www.beckhoff.com/en-en/support/download-finder/>
- Create a Beckhoff account
- Download **“TwinCAT Package Manager (TwinCAT 3.1 Build 4026)”**
- Open the downloaded installation file **“TwinCAT-Package-Manager-Setup.nnnn.exe”**
- Answer yes to the questions during the installation.

Install TwinCAT

- Start the newly installed “TwinCAT Package Manager”
- Fill in your User name (e-mail) and password for the account you created.
- Click save and answer Yes
- You only need Beckhoff Stable Feed
- The Outdated Feed can be necessary for connection to a physical system in the automation lab



Startup configuration

Feed configuration

Feed name: Beckhoff Stable Feed

Feed URL: <https://public.tcpkg.beckhoff-cloud.com/api/v1/feeds/stable/>

User name: anders.nilsson@hv.se

Password:

+ Save Cancel

TwinCAT Package Manager

Introduction
Welcome to the TwinCAT Package Manager

Feeds
Configure the feeds

Integration
Set the TwinCAT Engineering Integration

Feeds

The Beckhoff software products (workloads) including their components (packages) are available via so-called feeds.

The standard feeds are

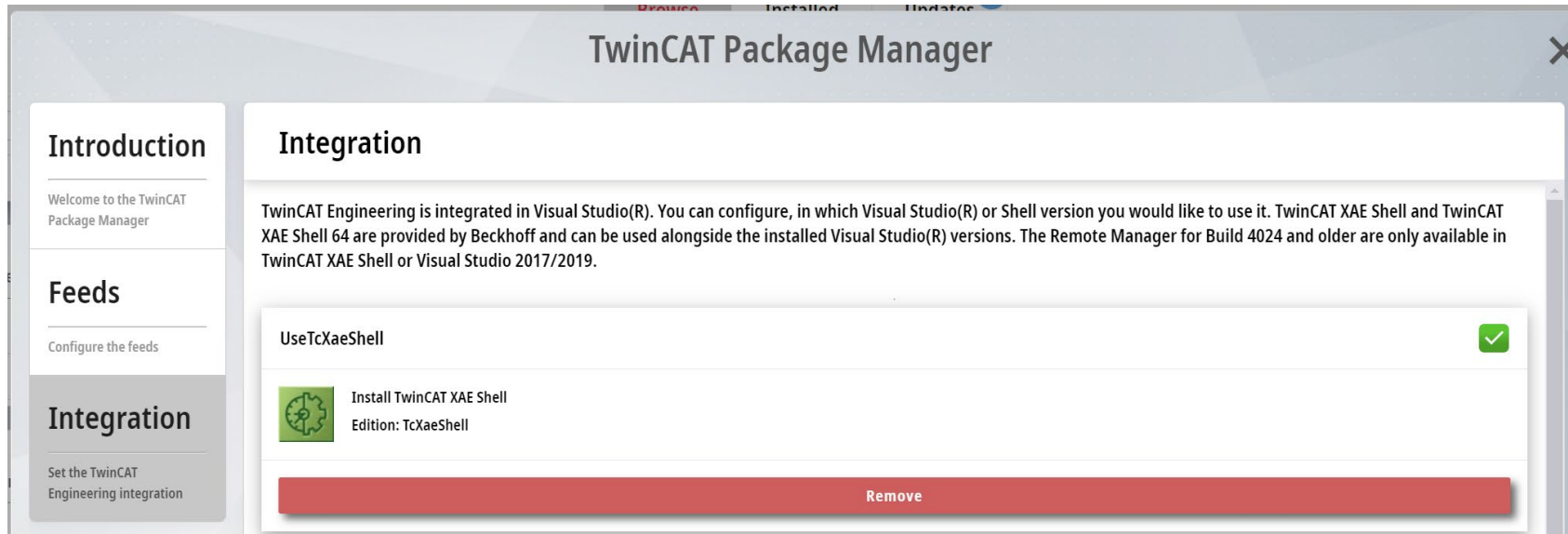
- The 'Beckhoff Stable Feed' is the default feed, which is the repository for officially released product versions.
- The 'Beckhoff Outdated Feed' provides access to software with an obsolete release version like old TwinCAT Standard Remote Manager versions.
- The 'Beckhoff Testing Feed' can be additionally configured to access test versions.

The login to all Beckhoff feeds is done via your myBeckhoff account from the Beckhoff homepage.
It is also possible to configure custom feeds, which can either be your own Nuget Package Server or a simple server folder path, where your defined Beckhoff and custom workloads and packages can be uploaded.

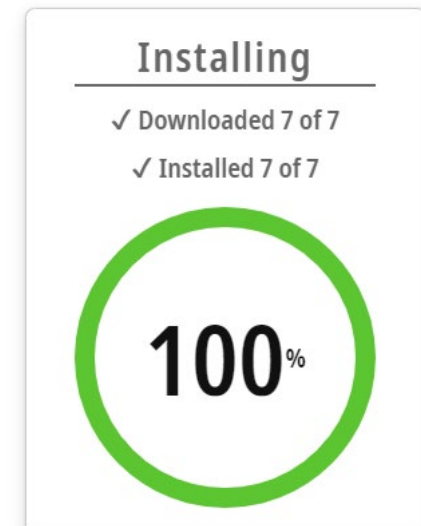
Priority	Enable	Feed name	Feed URL	
1	☒	Beckhoff Stable Feed	https://public.tcpkg.beckhoff-cloud.com/api/v1/feeds/stable/	↑ ↓ ✎ 🗑
2	☒	Beckhoff Outdated Feed	https://public.tcpkg.beckhoff-cloud.com/api/v1/feeds/outdated	↑ ↓ ✎ 🗑

+ Save ↶

Check UseTcXaeShell64 and click next

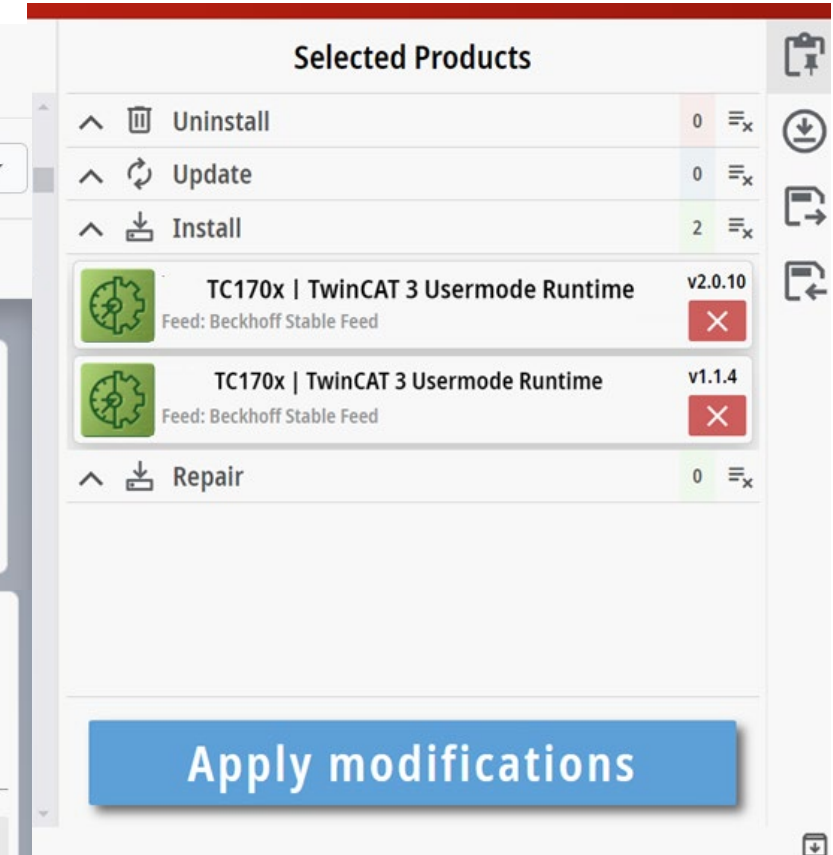
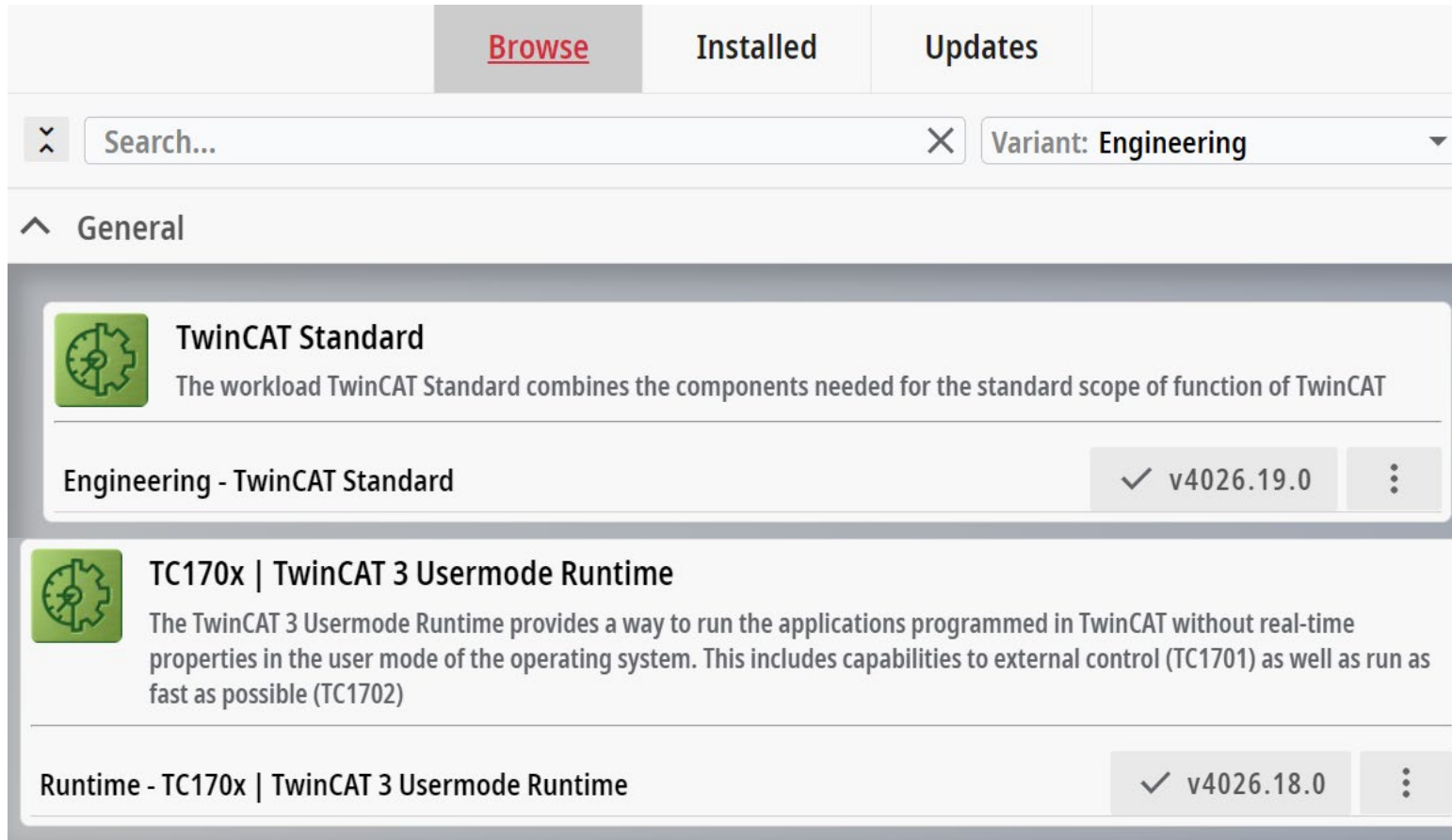


Click **Yes** to allow the installation program to install TwinCat

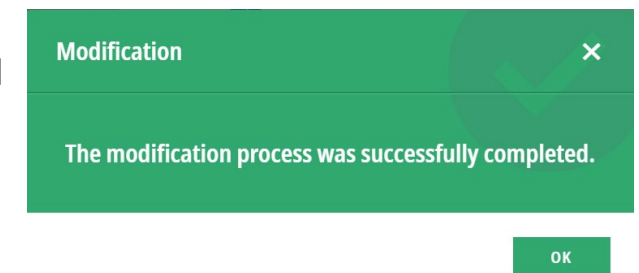


Have patience and wait until 100%

Install packages



1. Select: Engineering - TwinCAT Standard and TwinCAT 3 Usermode Runtime
2. Answer Yes to the question about making changes on your computer
3. Click on the pin symbol in the upper right corner and "Apply modification" to install the packages
4. When 100% and the completed sign appear, click OK and Close

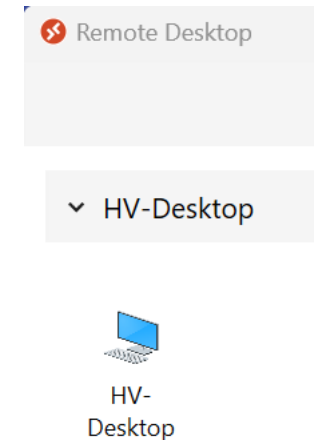


If you do not have access to a Windows Computer Install Remote Desktop

- Install HV-Desktop on your computer

<https://www.hv.se/student/it-support/datasalar-pa-distans/?sq=remote+desk>

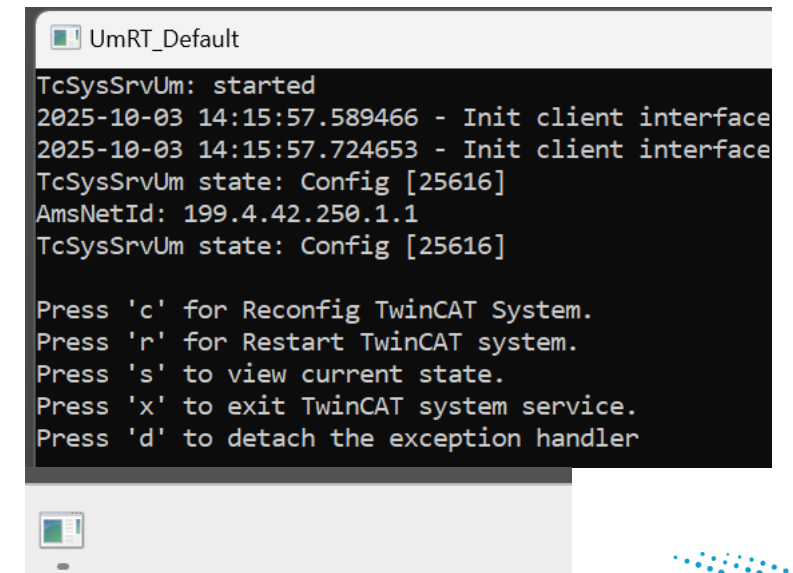
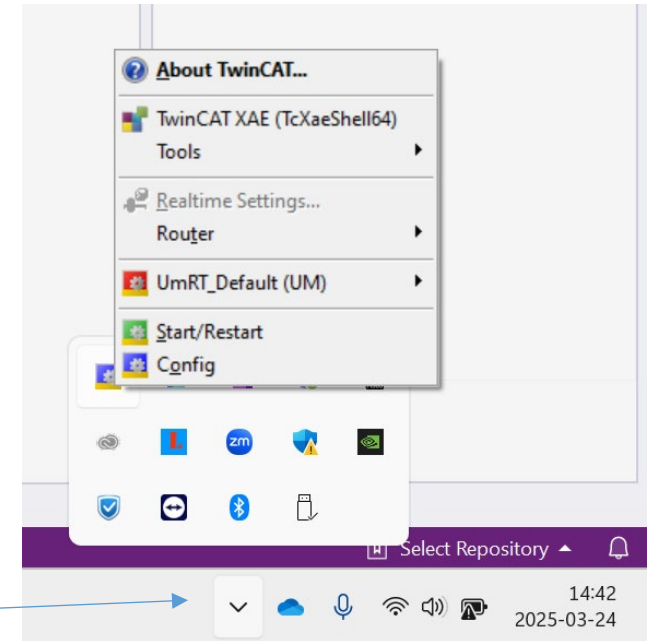
- Start HV-Desktop and run TwinCAT remotely.
- Download the simulation from Canvas
- Extract the simulation and run
- Don't forget to store the simulation
 - It is gone as soon as you close the connection.



UmRT_Default(UM)

When we run simulations of real devices, as we do in the assignments, we need to run a “runtime system” on our computer instead of a physical PLC. We will later download our applications to the runtime system to test run it.

- Start “UmRT_Default(UM)”
 - You will find it in the bottom right corner of the screen, under the ^ symbol and the blue cogwheel
 - Make sure that UmRt_Default has started
 - Check the upcoming DOS window 
 - If it does not look like the window here to the right, click on the blue cogwheel  Config and try again

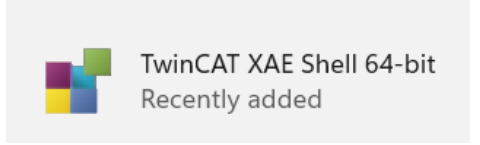
A screenshot of a DOS window titled 'UmRT_Default'. The window displays the following text:

```
TcSysSrvUm: started
2025-10-03 14:15:57.589466 - Init client interface
2025-10-03 14:15:57.724653 - Init client interface
TcSysSrvUm state: Config [25616]
AmsNetId: 199.4.42.250.1.1
TcSysSrvUm state: Config [25616]

Press 'c' for Reconfig TwinCAT System.
Press 'r' for Restart TwinCAT system.
Press 's' to view current state.
Press 'x' to exit TwinCAT system service.
Press 'd' to detach the exception handler
```

At the bottom of the window, there is a small icon of a DOS window.

Start a simulation in TwinCAT XAE Shell



- Download the wanted simulation from Canvas
- Open Windows Explorer 
- Right-click on the downloaded zip file and Extract All
- Click on the .sln file to start the project in TwinCAT

